

Contestant Number: _____

Time: _____

Rank: _____



PC SERVICING & TROUBLESHOOTING (305) **STATE 2021**

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 10 points each) _____ (500 points)

TOTAL POINTS _____ (500 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

1. The power supply of a computer supports all components within the computer system with which type of voltage?
 - a. lower voltage DC power
 - b. higher voltage DC power
 - c. lower voltage AC power
 - d. higher voltage AC power
2. The Northbridge chipset is connected directly to the CPU via what bus?
 - a. AGP Bus
 - b. Memory Bus
 - c. FSB
 - d. ATA Bus
3. What 15-pin power supply connector attaches to hard drives?
 - a. USB
 - b. SATA
 - c. eSATA
 - d. Thunderbolt
4. Which layer of the TCP/IP model makes up the top three layers of the OSI model?
 - a. Application Layer
 - b. Transport Layer
 - c. Network Access Layer
 - d. Internet Layer
5. When reassembling a computer, the technician notices that the thermal compound is dry and flaky. What should the technician do?
 - a. Scape off the old compound with a screwdriver and apply new compound.
 - b. Apply new compound over the old compound.
 - c. Clean off the old compound with alcohol wipes and apply new compound.
 - d. Continue to use the compound that is currently on the CPU.
6. What network protocol is used to announce network errors, network congestion and used for troubleshooting?
 - a. ICMP
 - b. FTP
 - c. Tracert
 - d. ipconfig /all
7. UEFI BIOS supports larger boot drives and includes features such as secure boot. What does secure boot prevent?
 - a. Prevents robotkits from accessing the network
 - b. Prevents robotkits from accessing the system
 - c. Prevents rootkits from accessing the network
 - d. Prevents rootkits from accessing the system

8. Acme Company is now utilizing virtual cloud applications. When the employees show up to work, they realize they are unable to run the applications. What is the issue?
 - a. POST is not recognizing the virtualization support.
 - b. Virtualization support is not enabled in the BIOS.
 - c. Device drivers are not supported.
 - d. Not enough memory to support virtualization.
9. CPU throttling techniques are used in laptops to ensure the _____.
 - a. enhanced performance of the CPU
 - b. CPU is running at its peak speed
 - c. CPU does not generate an abundance of heat to overheat the system
 - d. conservation of battery life by reducing the bus speed between CPU and memory
10. When installing a new CPU into a current system, what factors must the technician consider before installing?
 - a. Motherboard form factor and memory allocation
 - b. Motherboard form factor and video card
 - c. Motherboard form factor and hard drive type
 - d. Motherboard form factor and power supply watts
11. A technician has been asked to install a VDI Server in a growing company. What are the benefits of utilizing such a device?
 - a. Hypervisor segments servers into VM that in turn host standalone workstations.
 - b. Hypervisor segments servers into VM that in turn host virtual desktops.
 - c. All applications and data are stored on the workstations.
 - d. Servers are only used for client/server access.
12. When cleaning computer system fans, what tools should a technician use?
 - a. Compressed air spraying indirectly on the fans
 - b. Compressed air spraying directly over the fans
 - c. Dry lint-free cloths
 - d. None of the above
13. An end user has placed a data disk into the DVD drive. When they try to eject the disk, the tray does *not* open. What should the technician advise the user to do?
 - a. Insert a screwdriver and pry open the tray.
 - b. Insert a pin into the small hole on the front of the drive.
 - c. Remove the top of the drive enclosure.
 - d. Unplug the power to the drive, and press the eject button.
14. After upgrading the computer system with a new motherboard, CPU and adding memory, the user turns the system on. The system constantly reboots after booting into
 - a. Flash the BIOS.
 - b. CPU is causing the failure.
 - c. Memory mismatch errors.
 - d. Check the power supply to see if support the necessary watts needed to run the upgrades.

15. Hardware firewalls are used to block prevent unauthorized access to a LAN. What ports should be blocked to prevent access?
- UDP ports only
 - TCP ports only
 - STP ports only
 - UDP and TCP ports
16. The OSI and TCP/IP models are similar in theory. What layers make up the TCP/IP model?
- Application, session, network and data link
 - Application, transport, Internet and session
 - Application, transport, Internet and network access
 - Application, session, Internet and network access
17. Several users *cannot* access web pages that require login status. The network administrator inspected the firewall parameters and finds that HTTPS is being blocked. What port should the network administrator open?
- 22
 - 53
 - 80
 - 443
18. A computer technician is monitoring traffic on a network where users are complaining about slow network performance. The technician is curious as to whether the new network management software is causing the slowdown. Which port number would the technician be looking for in the captured packets for SNMP traffic?
- 161
 - 443
 - 67
 - 68
19. Company ABC is merging with another company. The size of the company will increase dramatically. In order to implement a new IP configuration, the network administrator needs to come up with a solution to address the entire network. What should the network administrator do?
- Manually configure all workstations using APIPA.
 - Configure all workstations using DHCP.
 - Use a DNS server to create a new IP address scheme.
 - Use ICMP to locate the network congestions.
20. When using IPv6 address schemes, IPv6 address can be compressed. Using the 2001:0db8:cafe:0100:0f0d:0000:0000:00AA/96, what is the best way to compress the address?
- 2001:0db8:cafe:0100:0f0d::00AA/96
 - 2001:db8:cafe:1:f0d::AA/96
 - 2001:db8:cafe:100:f0d::AA/96
 - 2001:0db8:cafe:0100:0f0d::AA/96

21. Yesterday, employees were able access all web services. When they came to work today, no one could access the web services. The network administrator uses the command prompt and notices that the IP Addresses all have 169.254.x.x. What should the network administrator check first?
- Turn on the device that is acting as a DHCP server.
 - Enable the client machine to receive an IPv4 address automatically.
 - Check the DNS server setting.
 - Reenter the IPv4 address on the DHCP server.
22. Devices such as printers and laptops need what interface to access SD Cards?
- M.2 card
 - Mini-PCIe adapter
 - Card reader
 - eSATA port
23. When printing on special paper that allows multiple copies to be printed, what type of printer will achieve this output?
- Inkjet printer
 - Impact printer
 - Laser printer
 - Thermal printer
24. Linux file systems use ext3 and ext4. What is the advantage of using ext4 over ext3?
- ext3 is a 32 bits file system while ext4 is a 64 bit file system.
 - ext4 does not support NFS making it more desirable.
 - ext4 delivers far superior performance than ext3.
 - Linux ext4 file system support FAT and FAT32 while ext3 only supports FAT.
25. A customer brings in their laser printer and complains about how the toner smears all over the printouts. When the technician inspects the printer, he notices excess toner inside the cartridge area. What should the technician use to clean the unit?
- A lint-free cloth
 - A damp cloth
 - A HEPA-capable vacuum
 - A handheld vacuum
26. What is the difference between cloud computing and a data center?
- A data center provides access to shared computing resources, whereas, a cloud is a private on-premise facility that stores and processes local data.
 - Cloud computing solutions are network resources and services provided to the clients through the use of services from data centers.
 - The data center makes use of more devices to process data.
 - Data centers require cloud computing, but cloud computing does not require data centers.

27. Which cloud computing service provides users access to computer resources that provide processing power, data storage and networking withing a data center?
- PaaS
 - SaaS
 - ITaaS
 - IaaS
28. A customer wants to transfer vital data as well as current computer settings when upgrading from an old Windows computer to a Windows operating system on a new computer. What two Windows utilities would be useful accomplish this task?
- Windows User Manager and Windows Easy Transfer
 - Windows Upgrade Advisor and User State Migration Tool
 - Windows User Manager and User Upgrade Assistant
 - Windows Easy Transfer and User State Migration Tool
29. A user examines his disk space and notices that his drive has large portion of the drive, which is unallocated disk space. The user wants to extend the primary partition with NTFS. What must the user do after the primary partition is extended to make it usable?
- Convert the disk type to dynamic.
 - Ensure that the disk type is basic.
 - Format the disk with FAT64 file system.
 - Partition the new space as a basic disk.
30. A network administrator is preparing to install computer systems using remote network installations. The system currently does *not* have an OS. What can the technician do to achieve this task?
- Use FTP to boot to the network.
 - Use PXE remote network installation.
 - Use RISC to communicate with the network client.
 - Use full format because a full format uses the faster FAT32 file system, whereas a quick format uses the slower NTFS file system.
31. A technician needs to install Windows 10 Pro on 10 PCs in a timely fashion. As the technician readies the PCs to connect to the network, the systems' fail to connect to the remote network service. What is the possible cause?
- The wrong network drivers are loaded in the image file.
 - The SID has not been changed in the image file.
 - The NIC cards on the new PCs are not PXE-enabled.
 - Sysprep was not used before building the image file.
32. When UEFI was introduced, it made available GPT which provided more flexibility in partitioning disks. What did the GPT replace?
- MBR
 - TFTP
 - STP
 - ext4

33. A company wishes to implement access controls that allows for two factor authentications. What two factors achieves this task?
- VPNs and VLANs
 - IDS and IPS
 - Passwords and fingerprints
 - Tokens and hashes
34. When implementing file encryption with Windows BitLocker, which two factors are needed to implement this solution?
- TPM and EFS
 - At least two volumes and password management
 - TPM and at least two volumes
 - EFS and only one volume
35. Which type of security threat falsifies the header of the email to make it look as if it is a legitimate corporate email?
- Phishing
 - Trojan
 - Malware
 - Spoofing
36. During company time an employee constantly is checking on personal business using the company laptop. The employee has worked there for many years. How should the network administrator handle the situation?
- Review the acceptable use policies with the employee.
 - Review the Incident handling policies.
 - Remove all network rights from the employee.
 - Do nothing as it is common practice.
37. The local security policy tool within Windows utilities allows network administrator to manage what tasks?
- View security logs to locate potential network security issues.
 - Configure password rules and account lockout policies on a system that is not part of a domain.
 - Configure active directory security tools.
 - Configure profiles and set disk space limitations.
38. When an employee prints documents to his local printer, he noticed that it is printing characters rather and texts. What should the technician advise the employee to do?
- The wrong or an outdated printer driver is installed; put in a ticket to have the technician update the driver.
 - Printer does not have enough memory; put in a request to add memory.
 - The printer cable is damaged; ask the technician to replace the cable.
 - The printer lid has not been closed securely.

39. Any type of network end device is considered a network host _____.
a. when it is connected to a workstation
b. when it is connected to a server
c. when it is connected to a switch
d. when it is connected to a VPN
40. In today's society, people are in control of their own personal network devices. Which technology is *most* often used to connect devices to a PAN?
a. IEEE802.11n wireless
b. Bluetooth
c. Fiber channel networks
d. IR
41. What technology allows users to connect seven devices simultaneously?
a. IR technology
b. Bluetooth technology
c. WiMax technology
d. ZigBee protocols
42. Which option is *not* a function of a proxy server?
a. Detecting malicious signatures
b. Firewall and network data filtering
c. Network connection sharing
d. Data caching
43. The IEEE 802.11n wireless standard operates at what frequency range?
a. 5 GHz and 2.4 GHz ranges
b. 2.4 GHz range
c. 5 GHz range
d. None of the above
44. Which Bluetooth data transfer rate is 25 Mbps?
a. Bluetooth 1.0
b. Bluetooth 2.0
c. Bluetooth 3.0
d. Bluetooth 4.0
45. What is the major difference between port triggering and port forwarding?
a. Port triggering can be used by any computer on your network, and you need to know the IP address of the specific computer.
b. Port triggering can be used by one computer on your network, and you need to know the IP address of the specific computer.
c. Port forwarding is passive and needs to be triggered.
d. Port triggering can be used by any computer on your network, and you need not know the IP address of the specific computer.

46. A technician is troubleshooting a DHCP network issue. Using the command prompt, what command will allow the technician to see if DHCP is enable?
- a. Ping
 - b. Ipconfig /dhcp
 - c. Ipconfig /all
 - d. Tracert
47. Which Windows desktop features resizes a window by dragging it to one of the edges?
- a. Snap
 - b. Shake
 - c. Peek
 - d. Slide
48. In Windows 10, the number of nested folders is limited to how many characters?
- a. 254
 - b. 255
 - c. 260
 - d. 300
49. Which IP address represents a class B address?
- a. 10.10.1.0
 - b. 130.25.1.0
 - c. 192.168.10.2
 - d. 244.25.0.0
50. Which IP address represents a private IP address?
- a. 12.10.10.0
 - b. 172.15.10.1
 - c. 192.168.10.2
 - d. 244.20.20.1